

ENVS 193DS Homework 03

Lucia Murillo

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```
# reading in packages
library(tidyverse)
library(here)
library(janitor)
```

```
library(readxl)
# creating new object from kelp data
kelp <- read_csv((here::here ("data","temp-kelp.csv")))
# creating new object from personal data
personal_data <- read_csv((here::here ("data","personal-data.csv"))) |>
  # cleaning names of columns
  clean_names() |>
  # dropping N/A values
  drop_na(date)
```

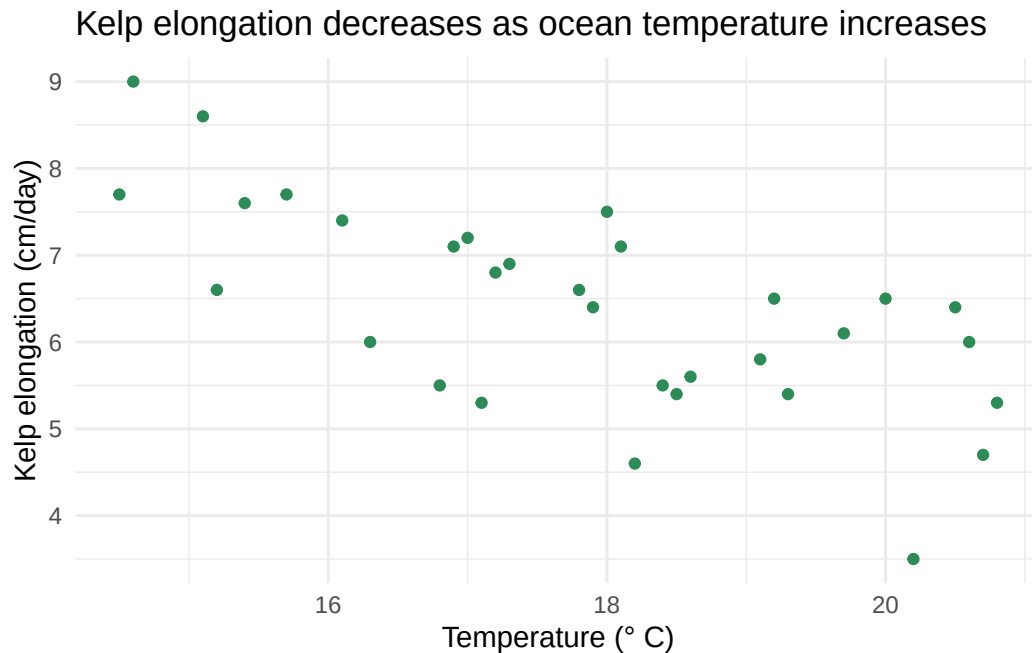
Problem 1: Giant Kelp fronds

(a)

One method of determining the strength of a relationship between temperature and giant kelp frond elongation is by finding the Pearson's correlation (r), which determines the direction and strength of the linear relationship between variables. The non parametric version is the Spearman rank correlation, which determines the monotonic relationship between variables.

(b)

```
# base layer ggplot
# data: kelp
ggplot(data = kelp,
       mapping = aes(
         # x-axis: temperature
         x = temp_c,
         # y-axis: kelp elongation
         y = kelp_elong)) +
# first layer: scatterplot
# changing color to royal blue
geom_point(col = "seagreen") +
# changing x and y axis titles, adding title
labs(x = "Temperature (\u00B0 C)",
     y = "Kelp elongation (cm/day)",
     title = "Kelp elongation decreases as ocean temperature increases") +
# changing to minimal theme
theme_minimal()
```



(c)

Checking assumptions:

In part 1b, I created a scatterplot to observe the patterns displayed in the `kelp` data. In this plot, a negative linear relationship between ocean temperature and kelp elongation is visible. This indicates that finding the Pearson's correlation would be the best way to determine the strength of this relationship since linearity is an assumption of this test. The other assumptions are met since the data is continuous and we can assume that the observations are independent.

Running a test

```
# correlation test
# variables: temperature + kelp elongation
cor.test(~ temp_c + kelp_elong,
  # data from kelp dataframe
  data = kelp,
  # pearson correlation
  method = "pearson")
```

Pearson's product-moment correlation

```
data: temp_c and kelp_elong
t = -5.189, df = 30, p-value = 1.366e-05
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.8359665 -0.4460157
sample estimates:
      cor
-0.6877489
```

(d)

To evaluate the strength of the relationship between ocean water temperature and kelp elongation, I used a Pearson's correlation test because the data is linear. I found a moderate negative (Pearson's $r = -0.687$, $t(30) = -5.19$, $p = 1.366e-05$, $0.05 = 0.05$) relationship between ocean temperature and kelp elongation. This leads me to reject the null hypothesis that the correlation is equal to 0.

(e)

I found that kelp elongation rates tend to decrease as temperatures increase. There is a moderate negative relationship between these two variables (Pearson's $r = -0.687$, $p = 1.366e-05$, $0.05 = 0.05$). In a time of rising temperatures worldwide, this is important to know as giant kelp is the base for many marine ecosystems.

(f)

```
# correlation test
# variables: temperature + kelp elongation
cor.test(~ temp_c + kelp_elong,
        # data from kelp dataframe
        data = kelp,
        # spearman correlation
        method = "spearman")
```

Spearman's rank correlation rho

```

data: temp_c and kelp_elong
S = 9216.1, p-value = 1.29e-05
alternative hypothesis: true rho is not equal to 0
sample estimates:
      rho
-0.6891684

```

These two tests have led me to the same interpretation of results. In the Spearman correlation test, I found a moderate negative relationship (Spearman's $\rho = 0.689$, $S = 9216.1$, $p = 1.29e-05$, $0.031 = 0.05$) between temperature and kelp elongation. This information leads me to reject the null hypothesis that ρ is equal to 0.

Problem 2: Personal Data

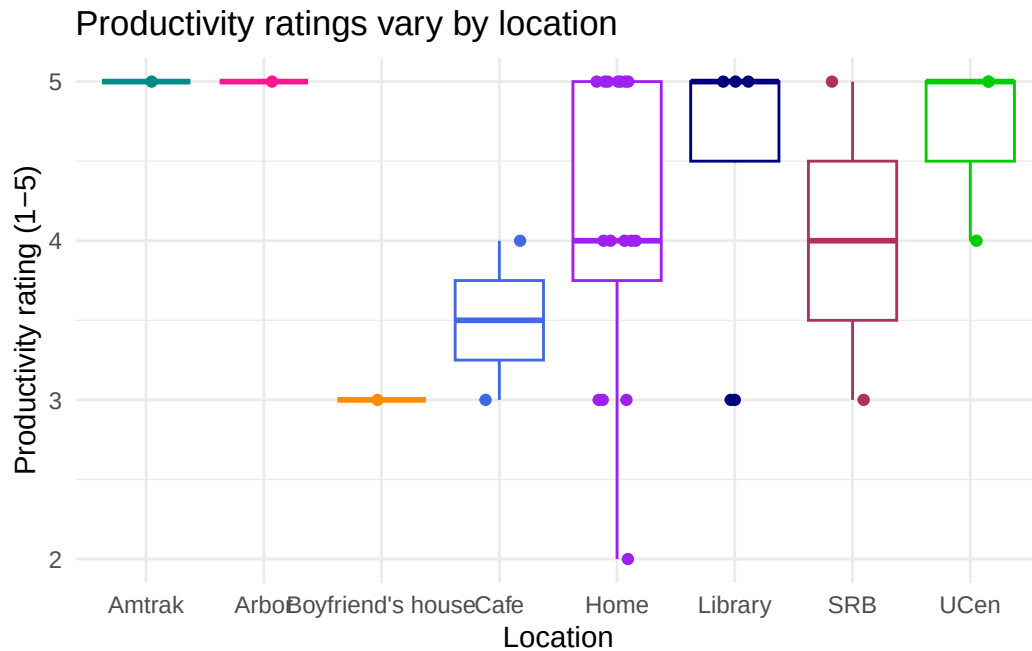
(a)

```

# ggplot base layer, data from personal_data
ggplot(data = personal_data,
      # x-axis: location
      # y-axis: productivity rating
      mapping = aes(x = location,
                    y = productivity_rating_1_5,
                    # color by location
                    color = location)) +

# boxplot layer 2
geom_boxplot() +
# jitter plot layer 3
geom_jitter(height = 0, width = 0.2) +
# changing titles of axes, adding title
labs(x = "Location",
      y = "Productivity rating (1-5)",
      title = "Productivity ratings vary by location") +
# adding minimal theme
theme_minimal() +
# changing colors of graph
scale_color_manual(values = c(
  "darkcyan", "deeppink", "darkorange", "royalblue", "purple", "navy", "maroon", "green3"))
# removing legend
theme(legend.position = "none")

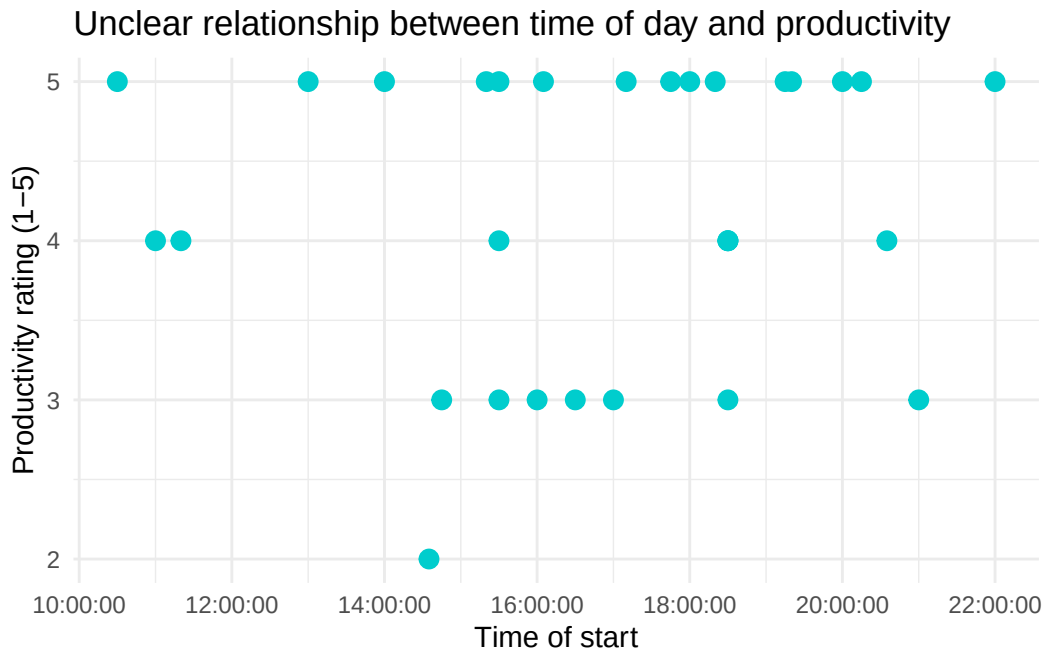
```



add subtitle, wrap x-axis labels?, take out grids and background

```
# ggplot base layer, data is personal_data
ggplot(data = personal_data,
      # x-axis: time of start
      # y-axis: productivity rating
      mapping = aes(x = time_of_start,
                    y = productivity_rating_1_5)) +

# add points
geom_point(
  # change color and size of points
  size = 3, color = "cyan3") +
#ADD geom_line(col = "cyan3") +??
# adding labels to axes and adding title
labs(x = "Time of start",
     y = "Productivity rating (1-5)",
     title = "Unclear relationship between time of day and productivity") +
# apply minimal theme
theme_minimal() +
# remove legend
theme(legend.position = "none")
```



add subtitle, change plot??, take out grid and background

(b)

Figure 1: Comparison of productivity levels at different locations. Each boxplot represents a location where I have studied, and each point represents a single study session.

Figure 2: The relationship between studying start time and productivity. Each point represents a single study session. **MAKE THAT ONE A LM??**

Problem 3: Affective visualization

(a)

One idea I had for an affective visualization of my data is making the time of day vs productivity graph be a gradient of time of day. The points could be stars on the nighttime end and little clouds or possibly butterflies or something similar on the day end. I would probably do this as a drawing either digitally or physically.

(b)

(c)

(d)

This shows my the changes in my productivity when studying when I start studying at different times of day. I chose to make it themed like a gradient of times of day. Each point is represented by a star or a cloud. This draft is a drawing done with colored pencils and pens. I was inspired by the Dear Data project by Stefanie Posavec and Giorgia Lupi as well as Jill Pelto's artwork.

Problem 4: Statistical critique

(a)

The test I have selected is the Mann-Whitney U-test. The response variable is treefrog density and the predictor variables are urbanization and crayfish presence. There is not a figure or table that is associated with the Mann-Whitney test that I am focusing on.

(b)

The test I am focusing on does not have a figure or a table.

```
knitr::include_graphics((here::here ("data","stream-screenshot.png")))
```


Table 5. Macroinvertebrate community indices in urban and natural streams in the Santa Monica Mountains and Simi Hills, California.

	2001	
	<i>urban</i> (n = 13)	<i>natural</i> (n = 20)
Taxonomic richness	23.15 ^a	29.40
EPT ^b taxa	5.08 ^a	9.40
Percent EPT invertebrates	23.26	32.98
Percent sensitive EPT (TV ^c = 0–2)	0.97 ^d	13.33
Percent intolerant (TV = 0–3) organisms	1.03 ^e	10.65
Percent tolerant (TV = 8–10) organisms	13.34 ^f	9.90
Percent most dominant taxon	45.91 ^e	33.69
Shannon diversity	1.65 ^d	2.23

^a Significant difference between urban and rural, low-gradient streams, bonferroni comparisons based on overall $p = 0.05$.

^b Aquatic insect orders Ephemeroptera, Plecoptera, and Trichoptera.

^c Tolerance values, a measure of sensitivity to disturbance and pollution with 0 being most sensitive and 10 most tolerant.

^d Significant difference between urban and rural streams at $p < 0.01$

^e Significant difference between urban and rural streams at $p < 0.05$.

^f Significant difference between urban and rural streams at $p < 0.10$.

This table shows the macro invertebrate community indices in urban and natural streams in the Santa Monica Mountains and Simi Hills, CA in 2001. It displays rows such as taxonomic richness and percent EPT (aquatic insect orders Ephemeroptera, Plecoptera, and Trichoptera) invertebrates, and compares urban and natural streams. Data points marked with different superscripts identify significant differences between the two at different p-values, bonferroni comparison, and other**.